Lecture 6

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Lecture 6

1. Reading Summary – Chapter 8
2. Reading Summary – Chapter 9
3. Association
4. Correlation and Covariance (Math)
5. Visually Estimating Correlations
6. Combining Datasets, Subsetting Datasets, and Ecological Correlation
7. It's a correlated, correlated world

Start reading Chapters 10-12 for next lecture.

Quiz 3 is Wednesday in lecture.

Section 8.1: The Scatter Diagram

- Introduction to scatter plots
- **Association**: There is an association between two variables if knowing one variable helps you predict the other.

Section 8.2: The Correlation Coefficient

- **Point of averages**: Center of cloud
- **Correlation Coefficient**: Measure of linear association (clustering around a line)
- Correlation is a number between -1 and 1
- Figure 6: Shows different correlations

Section 8.3: The SD Line.

- The SD line is the line around which the points visually cluster in a perfect world.
- When x increases by 1 SD, y increases by 1 SD. The slope is $SD(y)/SD(x)$.
- Not commonly used.

Section 8.4: Computing the Correlation Coefficient

- Convert the values in each list to standard units. Then compute the products and take the average.
- To see why correlation measures linear association, think about quadrants of the graph.
- Points in the first and third quadrant contribute positively to the correlation.
- Points in the second and fourth contribute negatively.

Reading Summary – Chapter 9

Section 9.1: Features of the Correlation Coefficient

- Correlation is not affected by changes in scale
- Adding a constant to one of the variables will not change the correlation coefficient.
- Multiplying one of the variables by a positive constant will not change the correlation coefficient.

Section 9.2: Changing SDs

- To estimate the correlation visually, you must mentally re-scale the plot so the standard deviations of x and y match up.

Section 9.3: Some Exceptional Cases

- The correlation coefficient is mainly useful for football-shaped data.
- Outliers can make correlation misleading.
- Correlation only measures linear association, not association in general.

Section 9.4: Ecological Correlations

- **Ecological correlations** are correlations based on rates or averages.
- Ecological correlations can overstate the strength of an association.

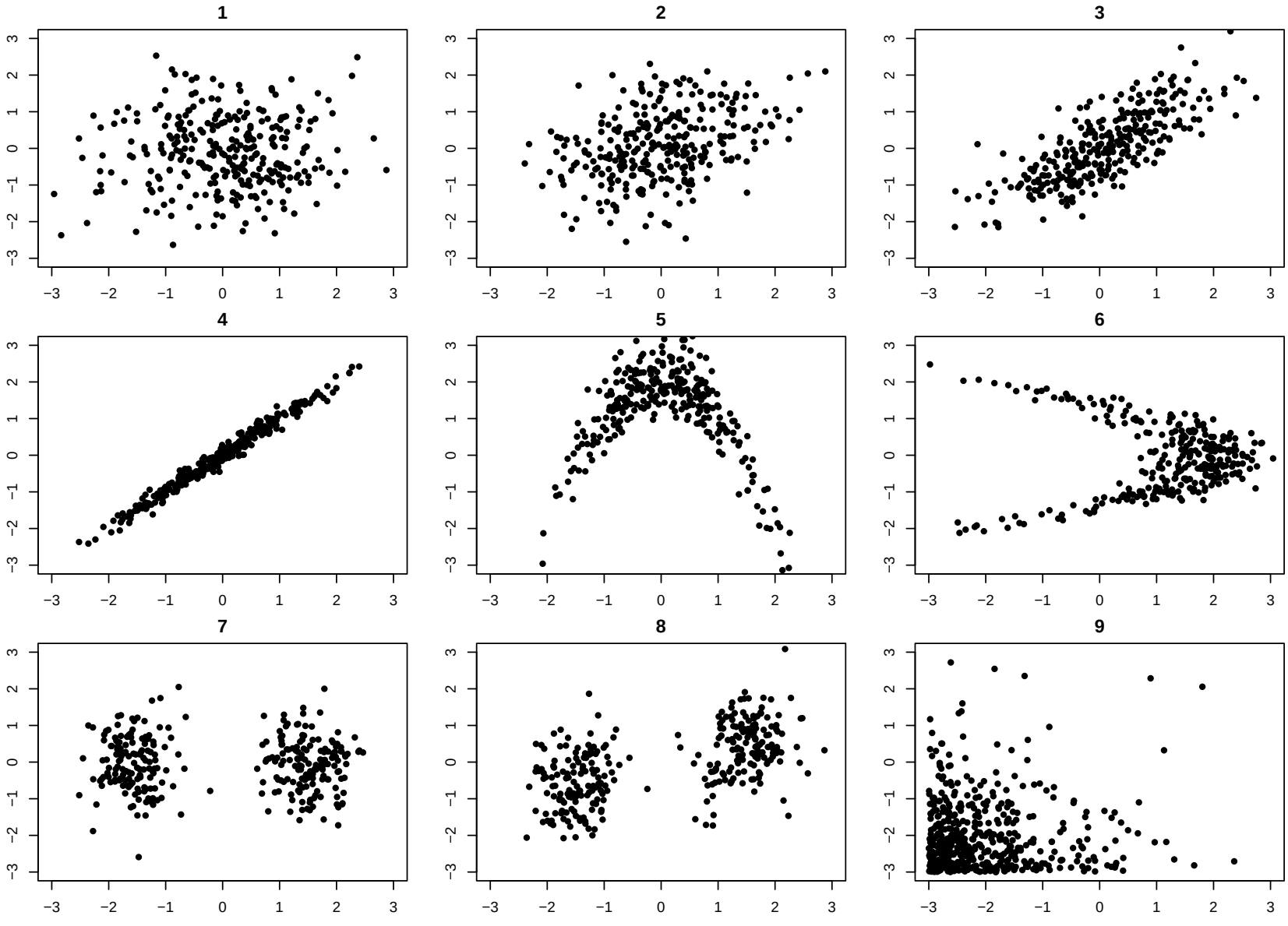
Section 9.5: Association is not Causation

- Correlation only measures association, and association is not causation.
- Associations can be the result of confounding.

Association

- There is an association between two variables if knowing one of the variables can help you predict the other.
- In some of the following plots, x and y are associated, and in others not. Which are which?

Plots



Correlation and Covariance (Math)

Correlation

- Suppose there are two lists of numbers of the same length: $x_1, x_2, \ldots, x_n$ and $y_1, y_2, \ldots, y_n$.
  - Define the **correlation** as

$$corr(x, y) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{SD(x)} \right) \left(\frac{y_i - \bar{y}}{SD(y)} \right)$$

  - Or in other words:

$$corr(x, y) = \overline{(x \text{ in SU}) \times (y \text{ in SU})}$$

Covariance

- We can rewrite the correlation as:

$$\frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{SD(x) \times SD(y)}$$

  - The quantity in the numerator,

$$\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}),$$

    is the **covariance** of x and y, written $cov(x, y)$

Correlation and Covariance

- Thus, the relationship between correlation and covariance is:

$$\text{correlation} = \frac{\text{covariance}}{SD(x) \times SD(y)}$$

- Or alternatively,

$$\text{covariance} = \text{correlation} \times SD(x) \times SD(y)$$

Correlation vs Covariance

- Covariance turns out to be very useful theoretically.
- However, we won't use it much in this class.
- Covariance is not a good measure of association.
  - Just like variance is not a good measure of spread.
- Correlation, however, is generally a good measure of linear association.

Correlation as a Measure of Linear Association

- Correlation is often a good measure of the strength of linear association between x and y.

1. Correlation is unaffected by changes of scale.
   - If you add a constant to every x number, the correlation won't change.
   - If you multiply every x number by a positive constant, the correlation won't change.
   - The same is true for y.
   - What happens if you multiply x or y by a negative constant?

Why is Correlation a Good Measure of Linear Association?

2. Correlation is always between -1 and 1.
   - A correlation of 1 means perfect positive linear association.
   - A correlation of -1 means perfect negative linear association.
   - A correlation of 0 means no linear association.

- Note: The value of the correlation doesn't have an easy interpretation.

Rules of Thumb for Correlation

- Different fields have different rules of thumb.
  - Engineers have a stricter interpretation than social scientists.
  - One rule of thumb (more social science):

Range of $\lvert correlation \rvert$	Interpretation
$0 - 0.3$	weak
$0.3 - 0.7$	moderate
$0.7 - 1$	strong

Sidenote: Alternative formula for covariance

- Using some algebra (not shown here), we can show that the covariance

$$\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

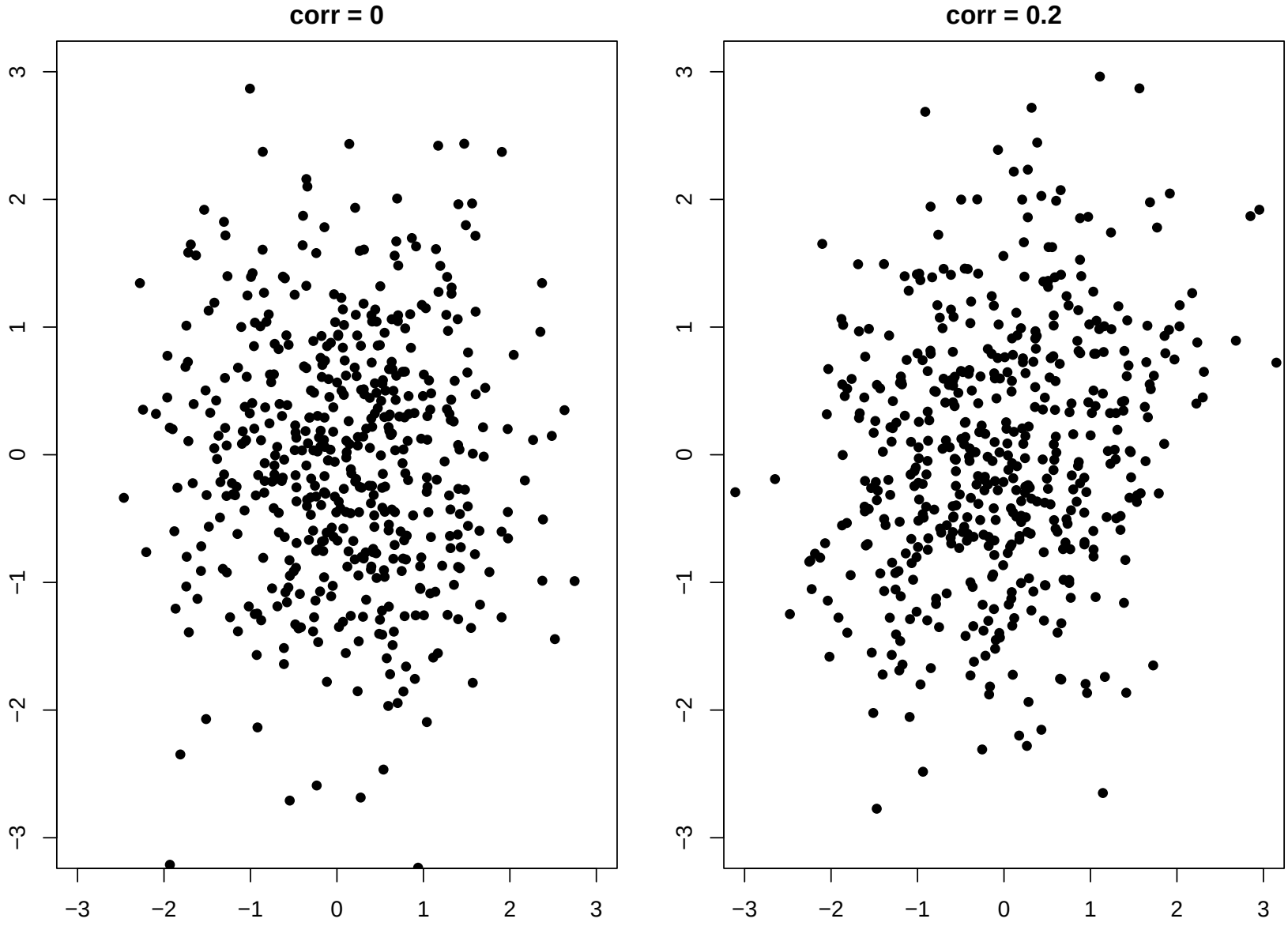
  is equal to

$$\overline{xy} - \bar{x}\bar{y}$$

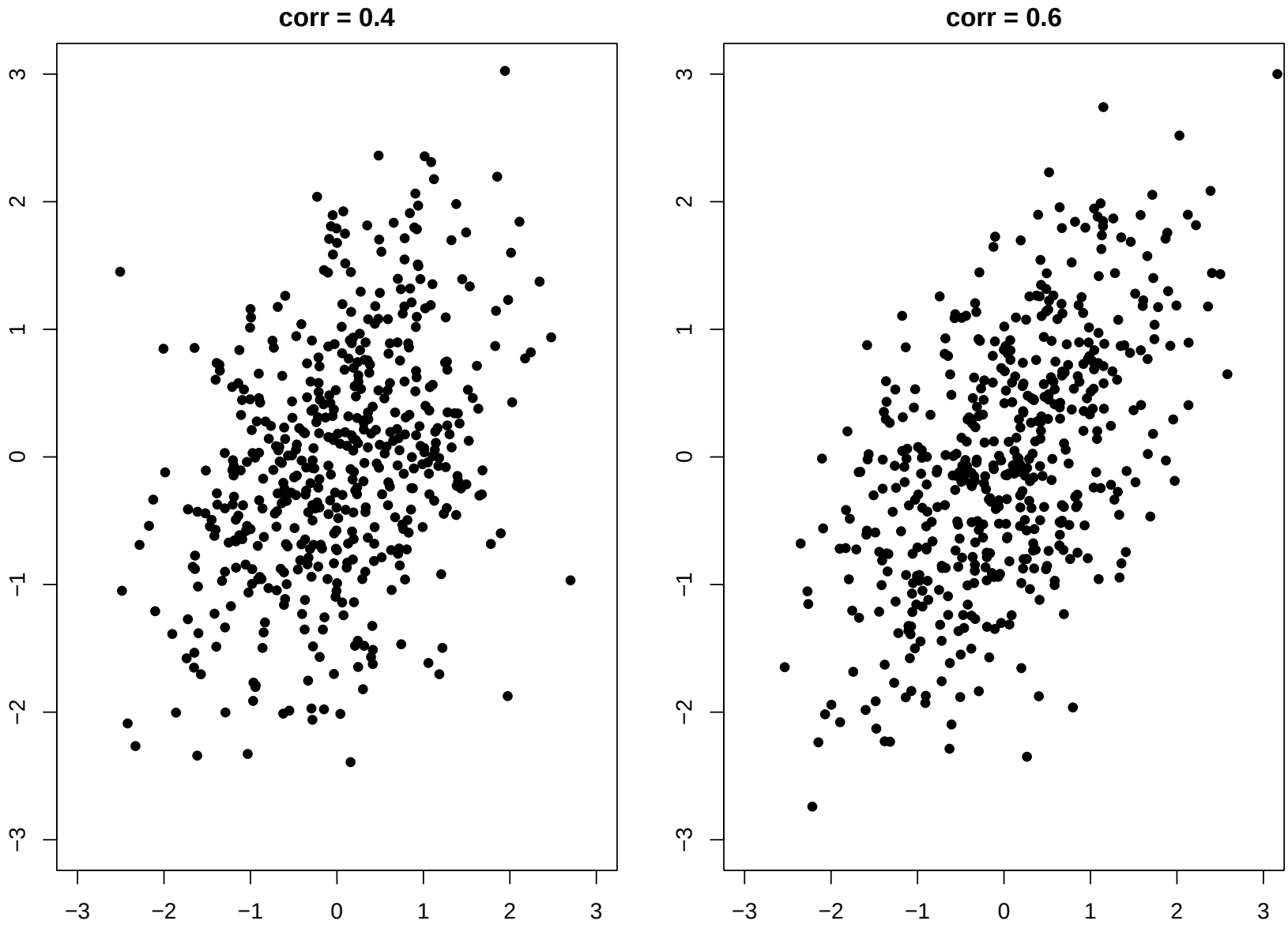
  - In words, this is the average of the products minus the product of the averages.
- Note that the variance of x is the covariance of x with itself.

Visually Estimating Correlations

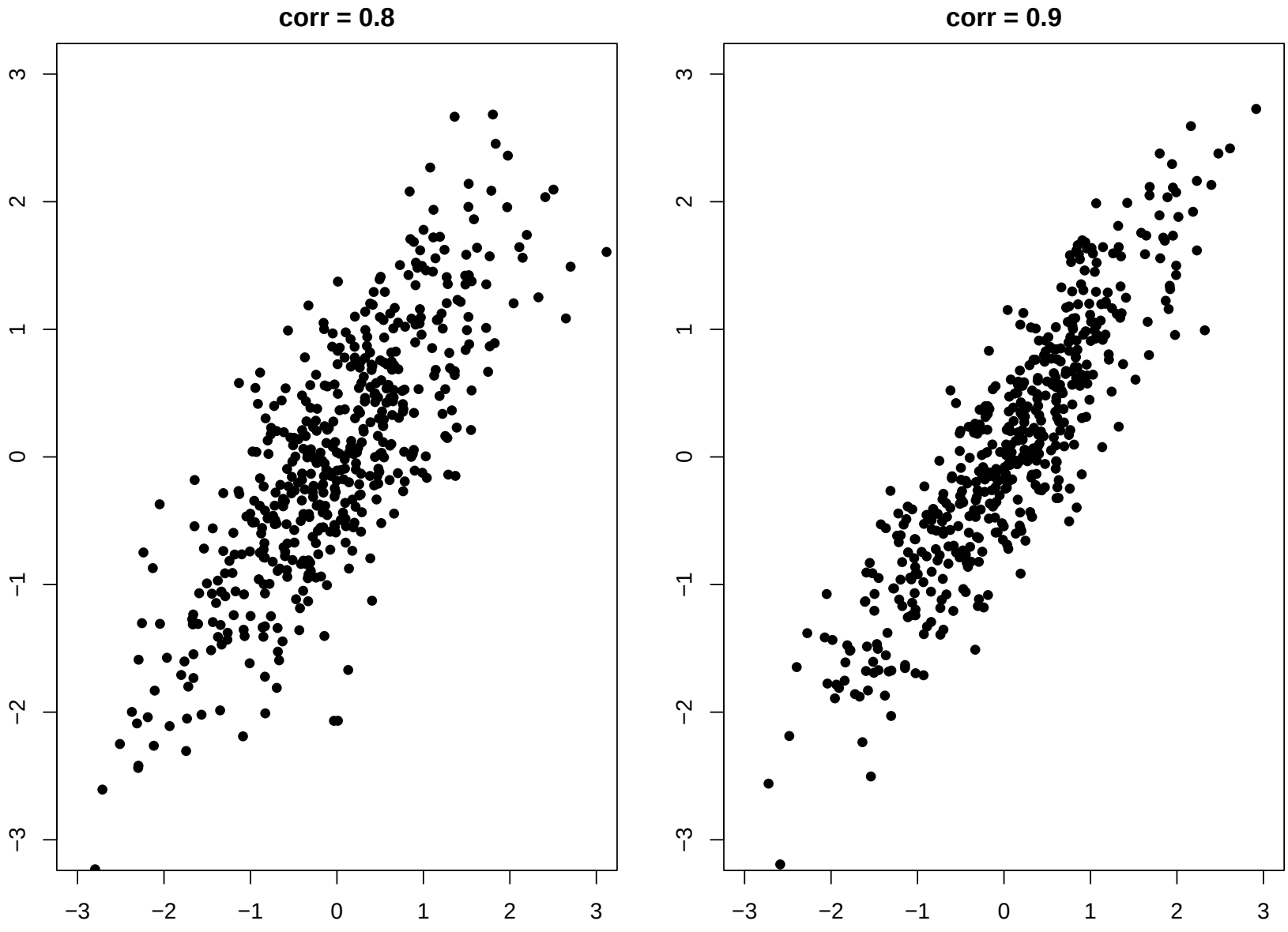
Visually Estimating Correlations 1



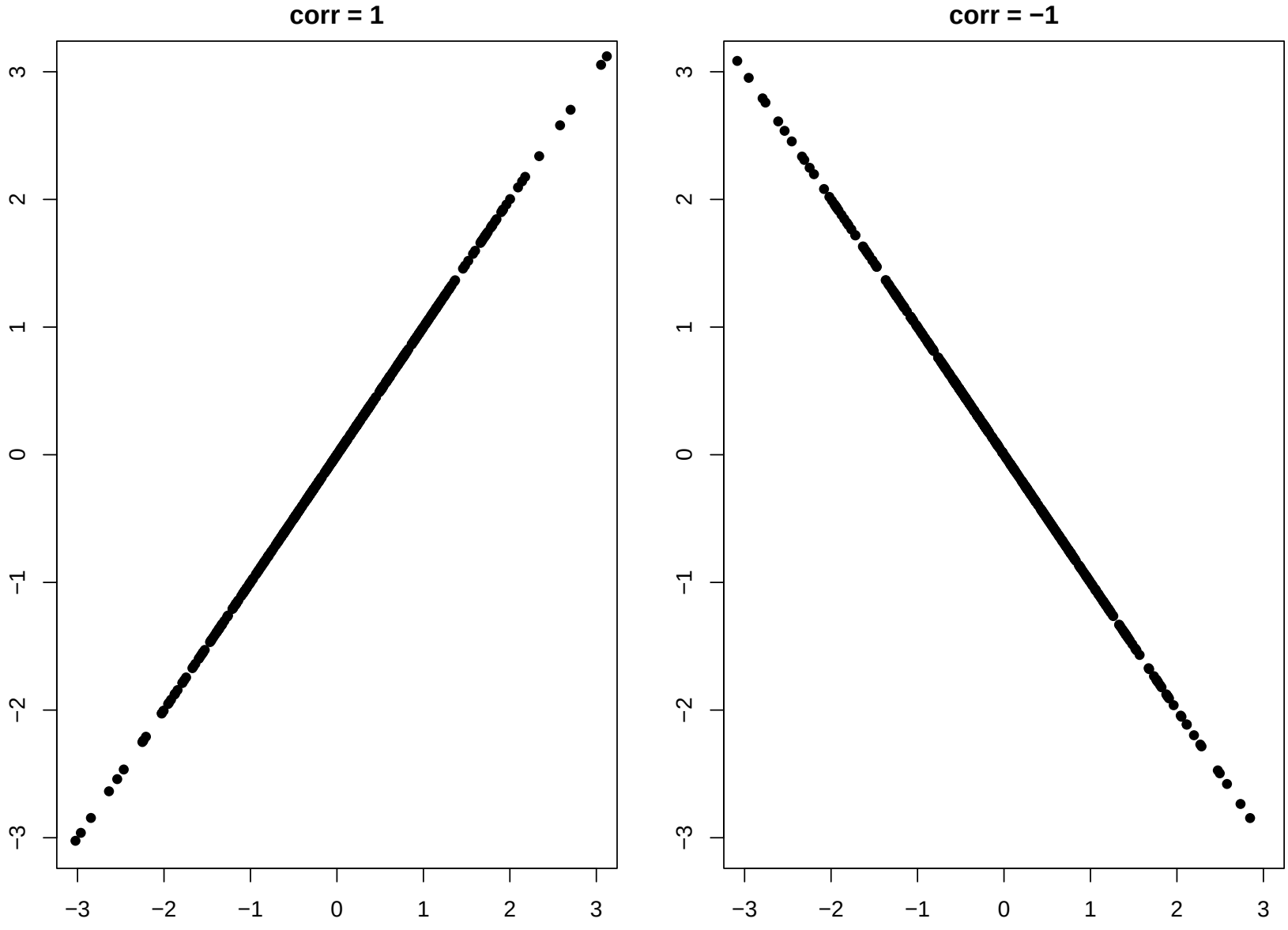
Visually Estimating Correlations 2



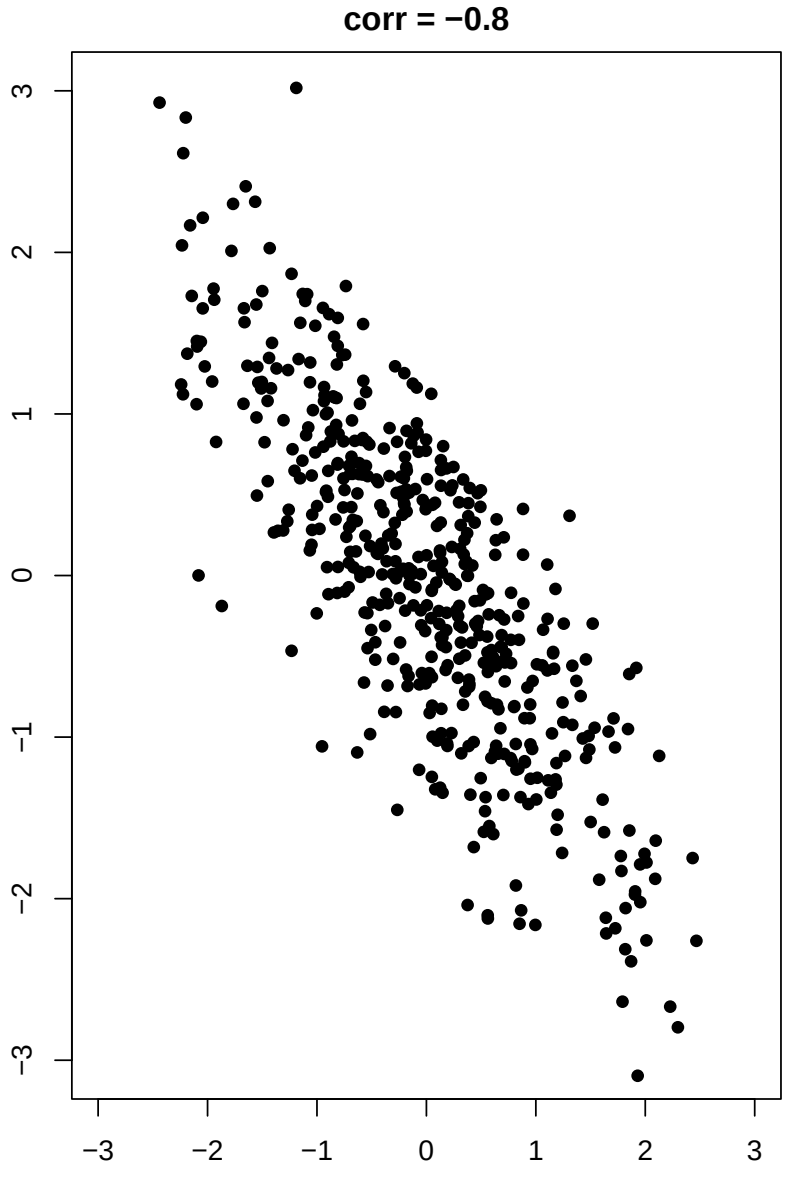
Visually Estimating Correlations 3

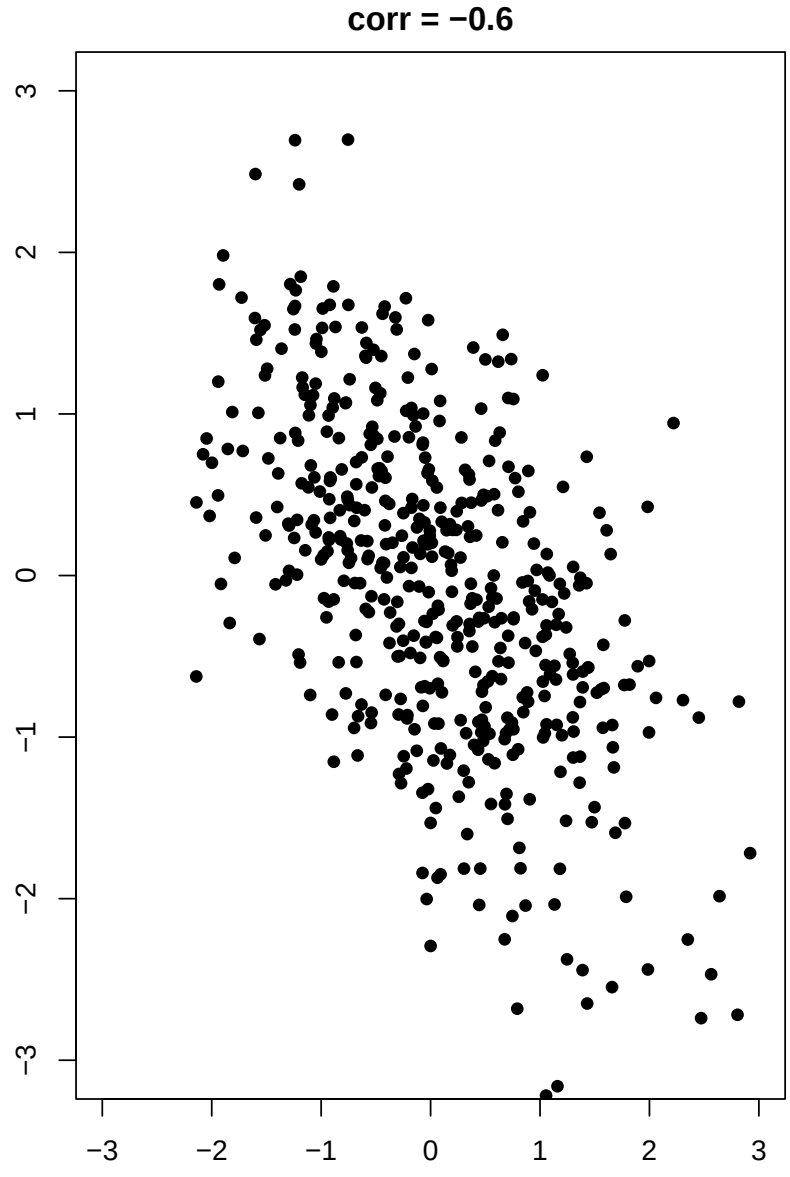


Visually Estimating Correlations 4

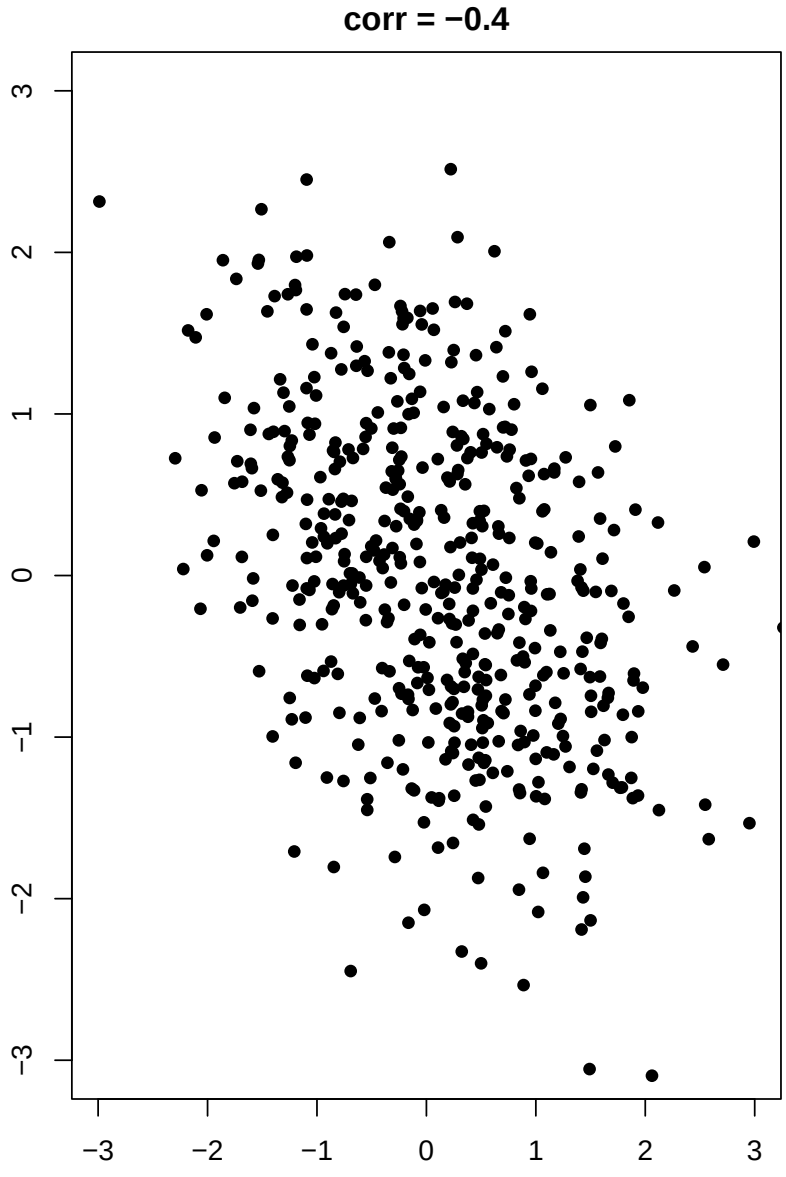


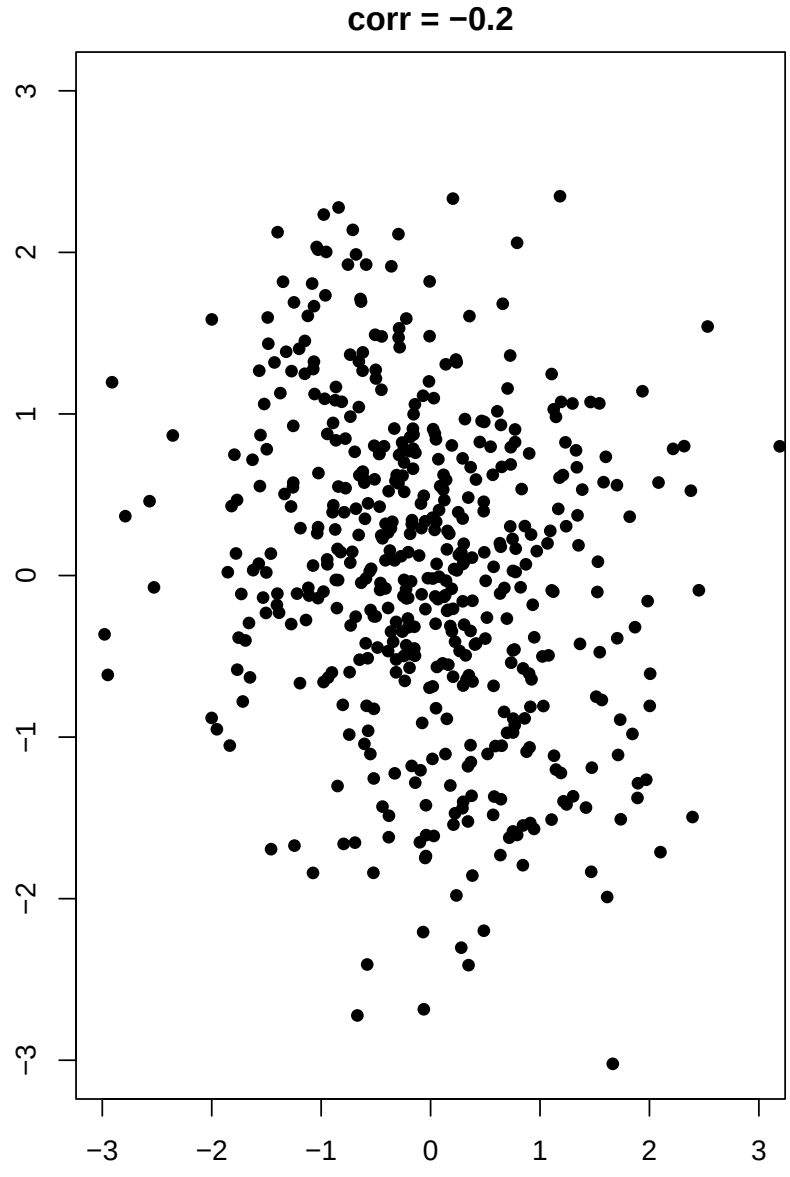
Visually Estimating Correlations 5





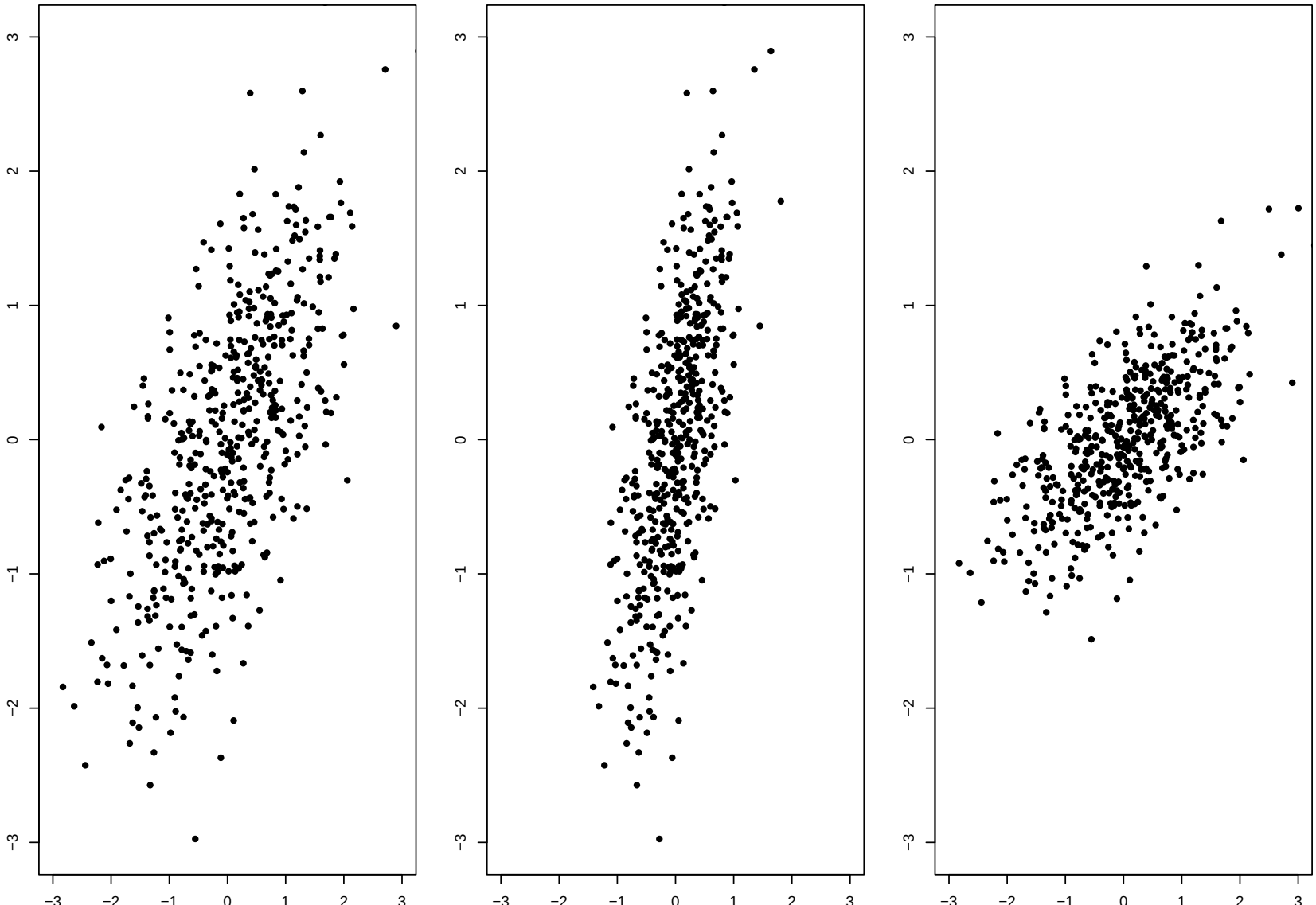
Visually Estimating Correlations 6





Watch Out for Scaling!

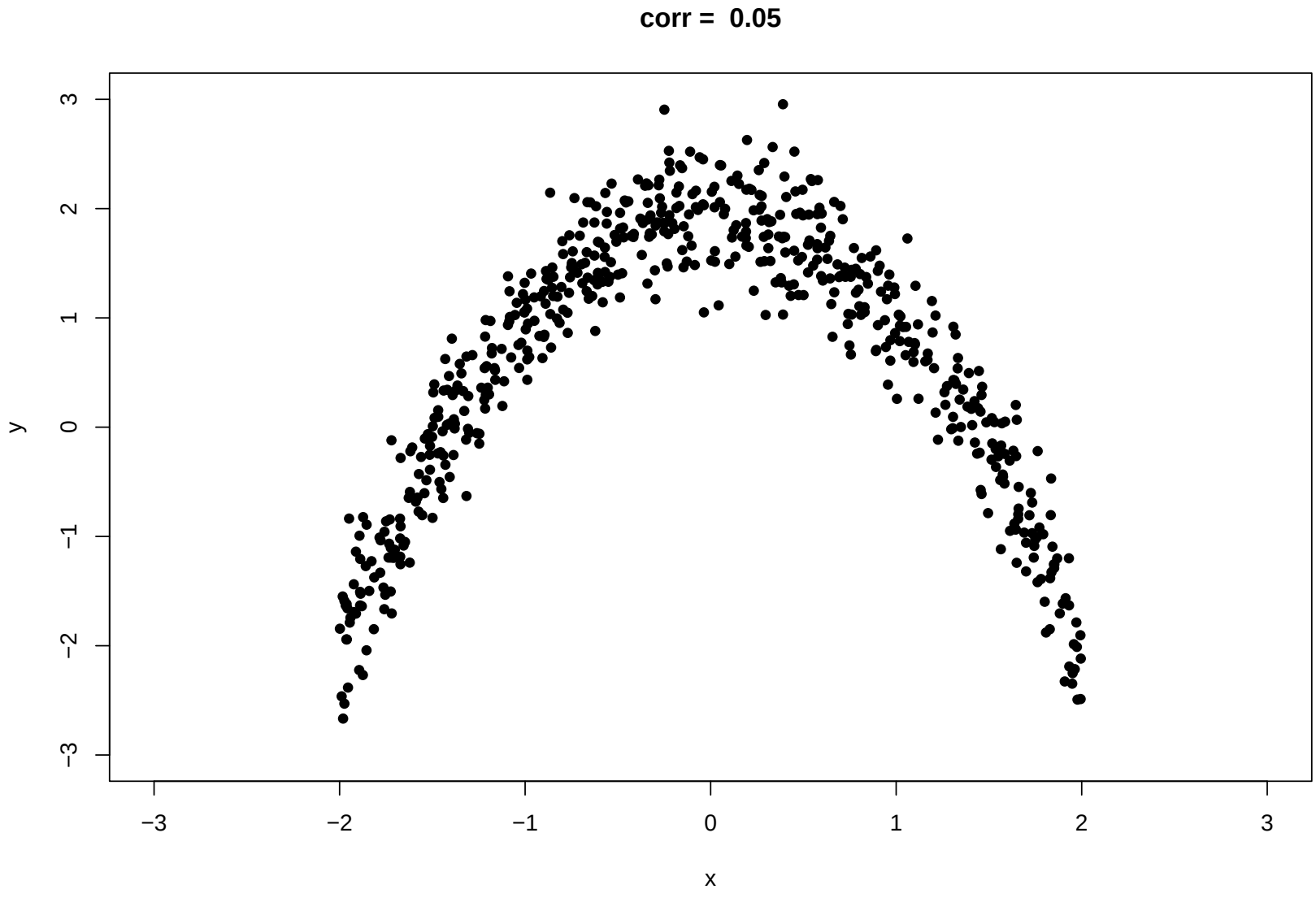
- All of the following plots have correlation of about 0.7.
- Rescale x and y to have the same standard deviation.



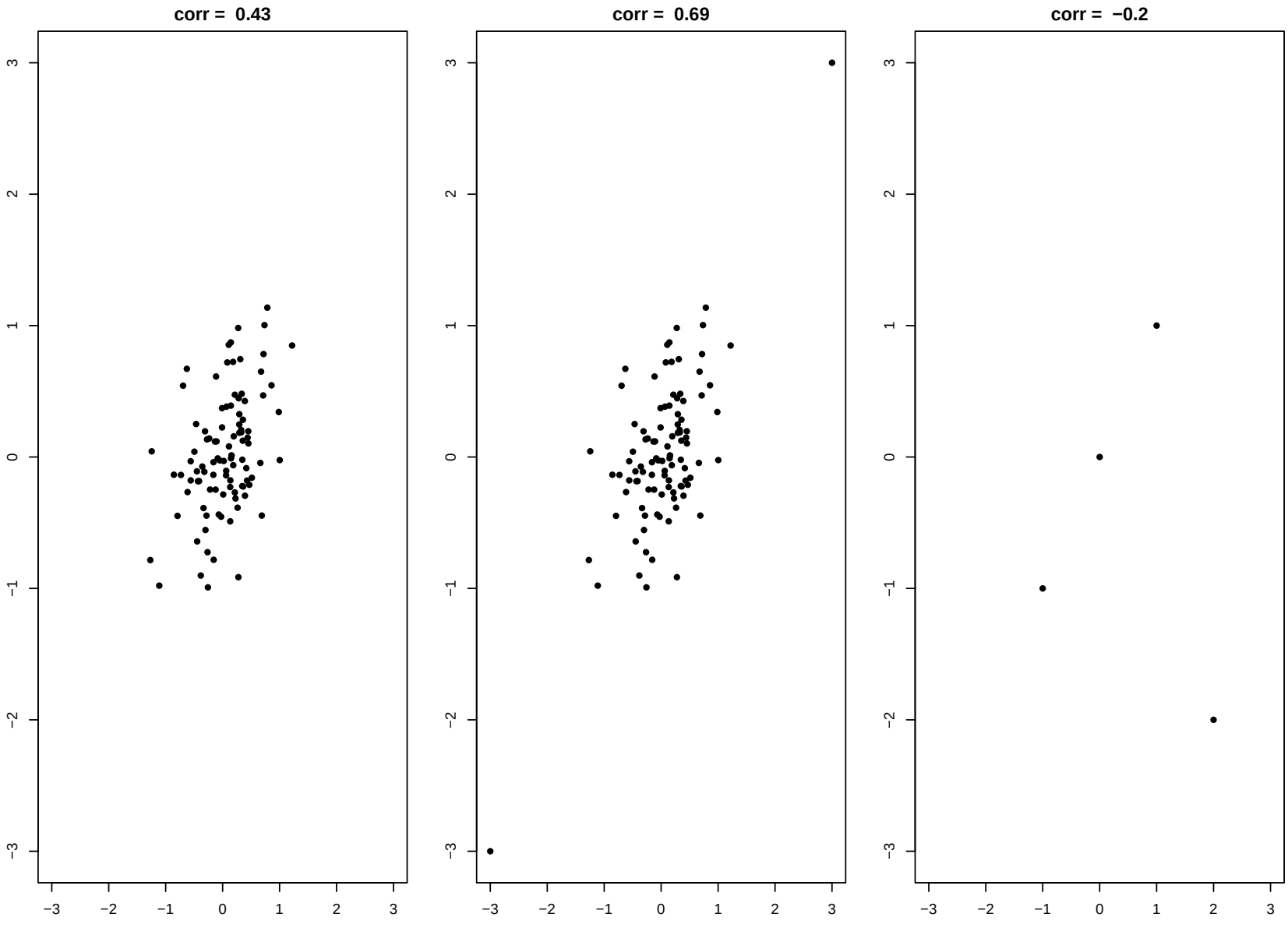
Non-Football-Shaped Data

- The correlation that we've defined is best for assessing linear relationships with football-shaped data.
- Correlation is not good at assessing non-linear relationships or relationships with non-football-shaped data.
  - Outliers
  - Clustering

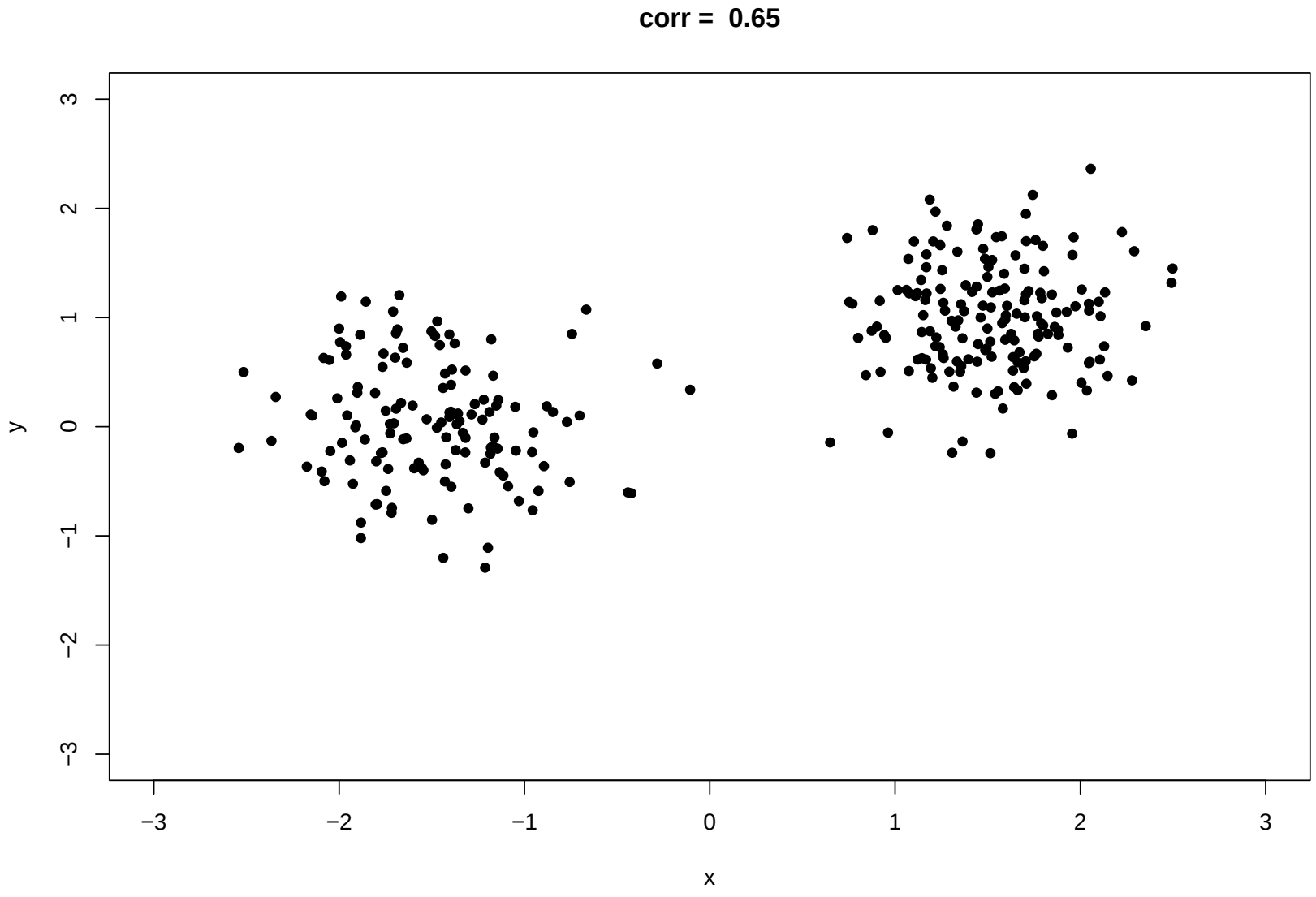
Nonlinear Data



Outliers



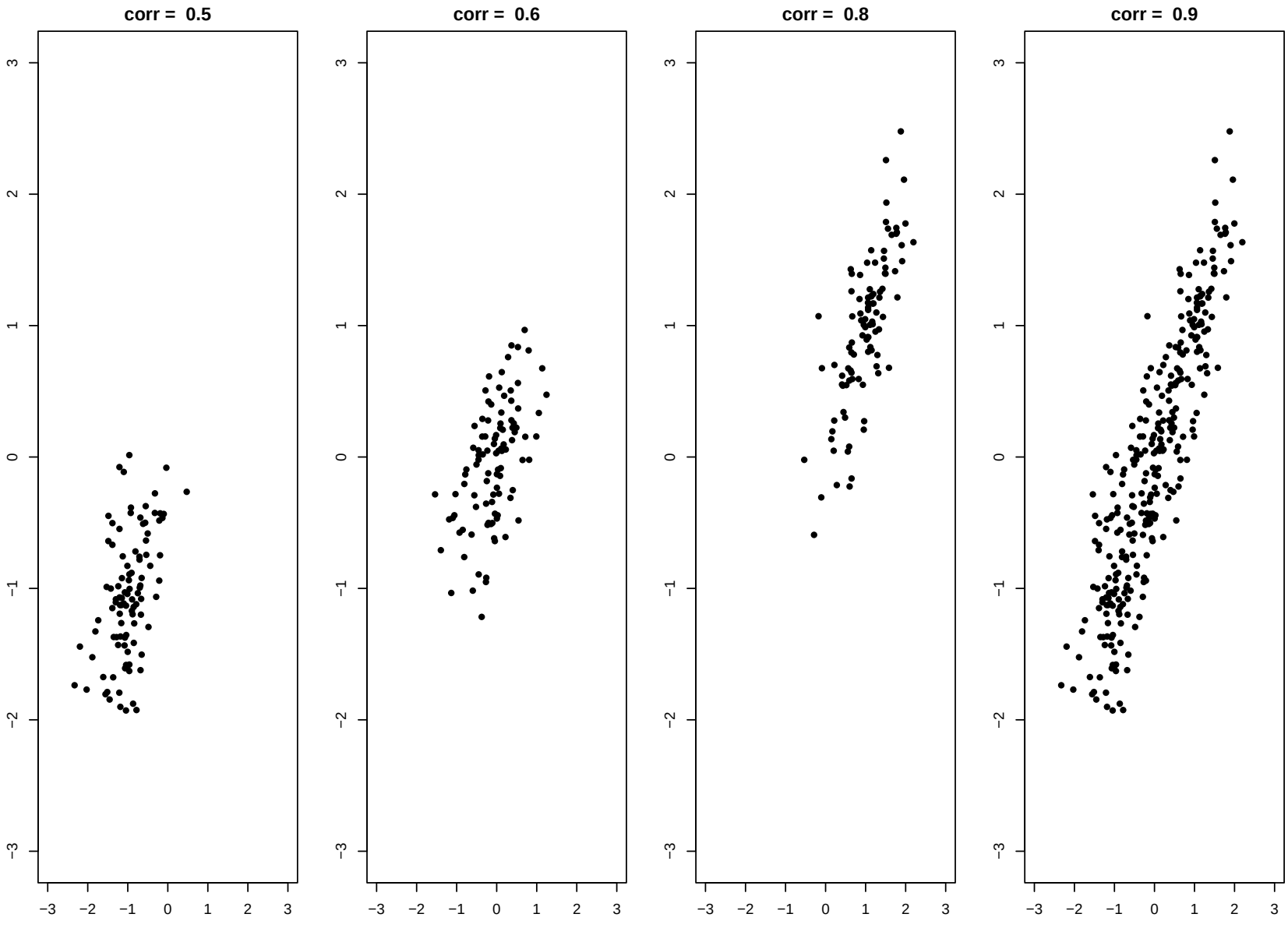
Clusters



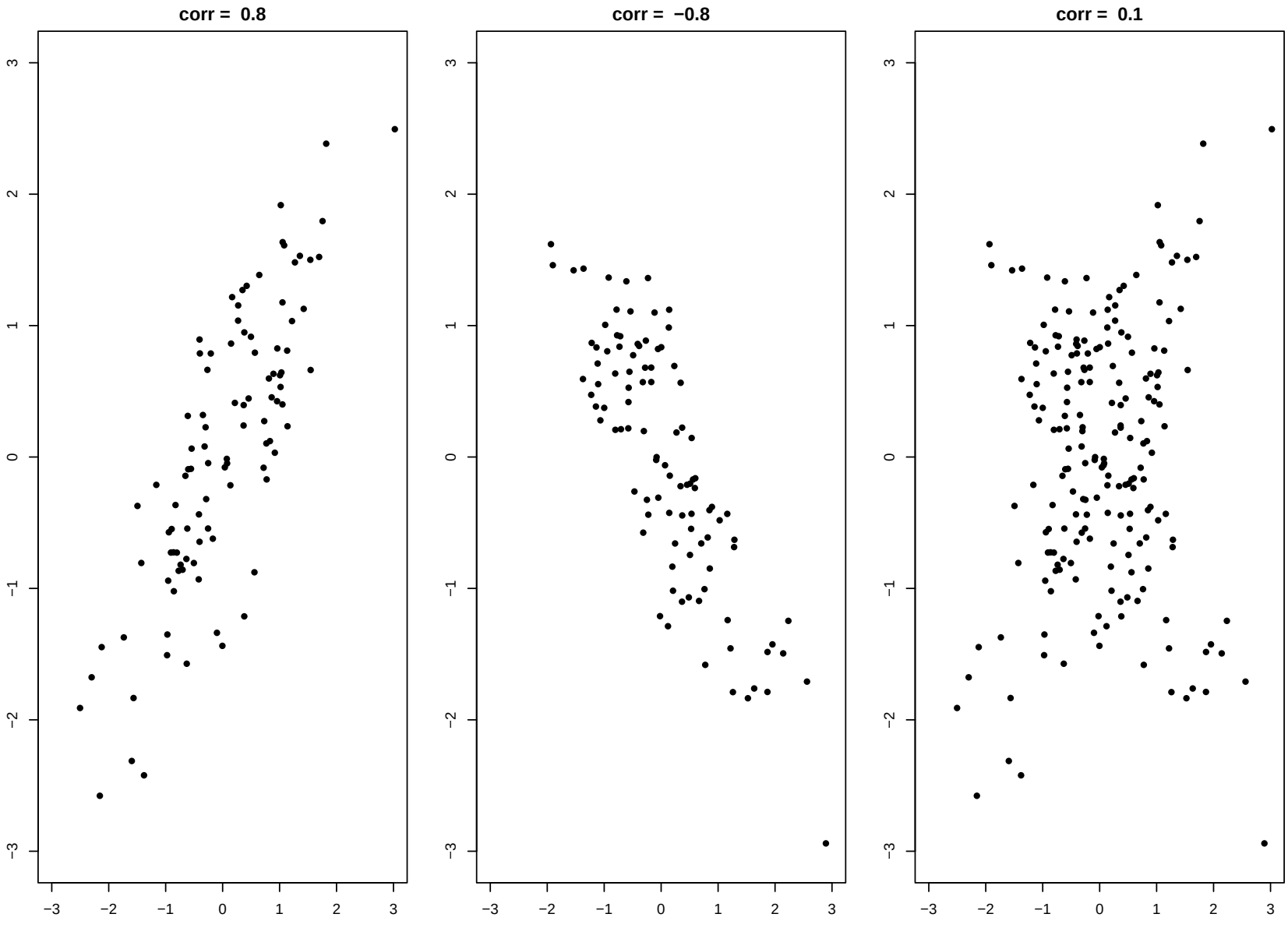
Combining Datasets, Subsetting Datasets, and Ecological Correlation

- Surprising things can happen to the correlation when you:
  - combine datasets,
  - take a subset of a dataset, or
  - look at averages or rates.
- This will be illustrated with artificial and real-life examples.

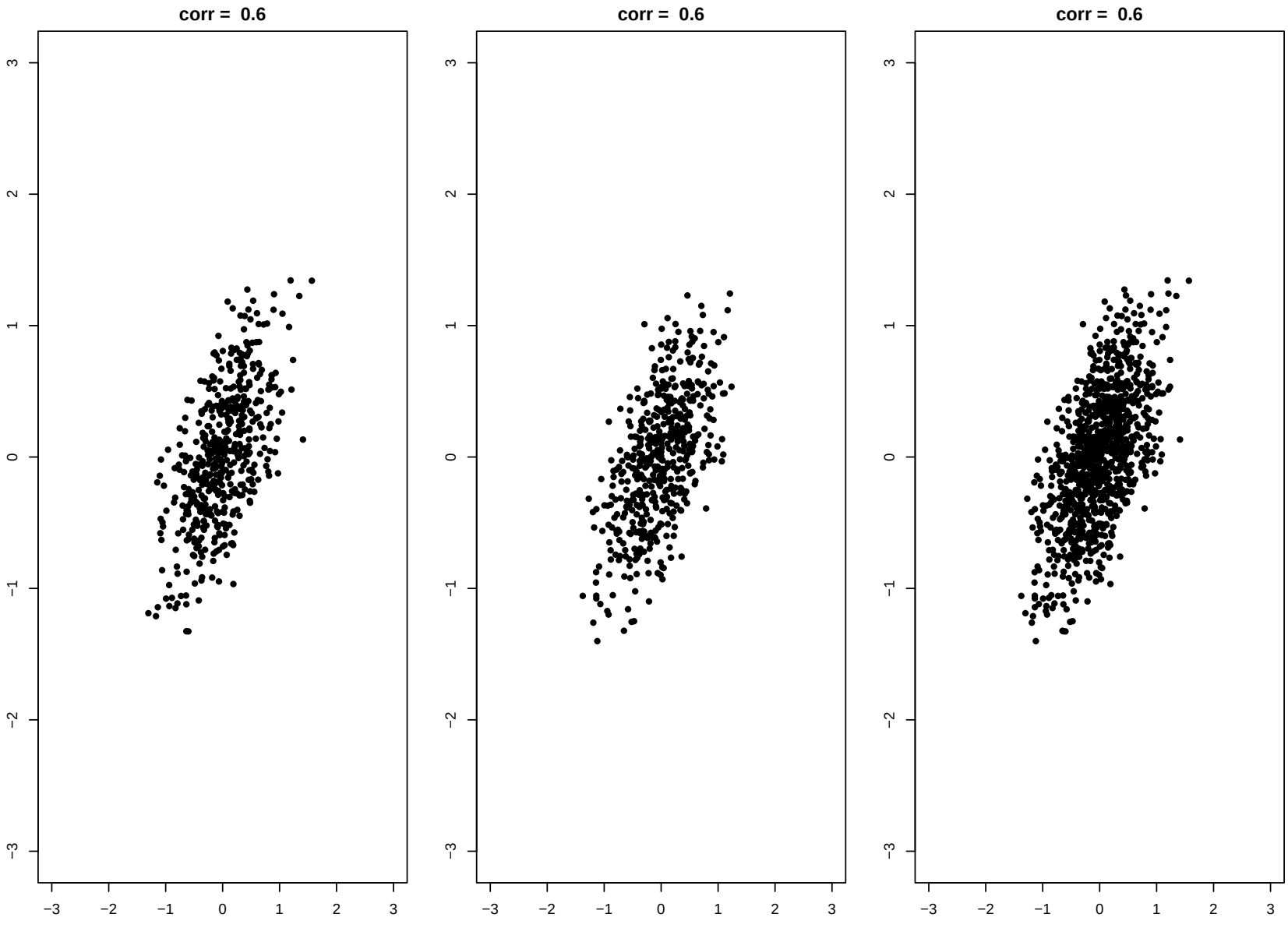
Combining Datasets 1



Combining Datasets 2



Combining Datasets 3

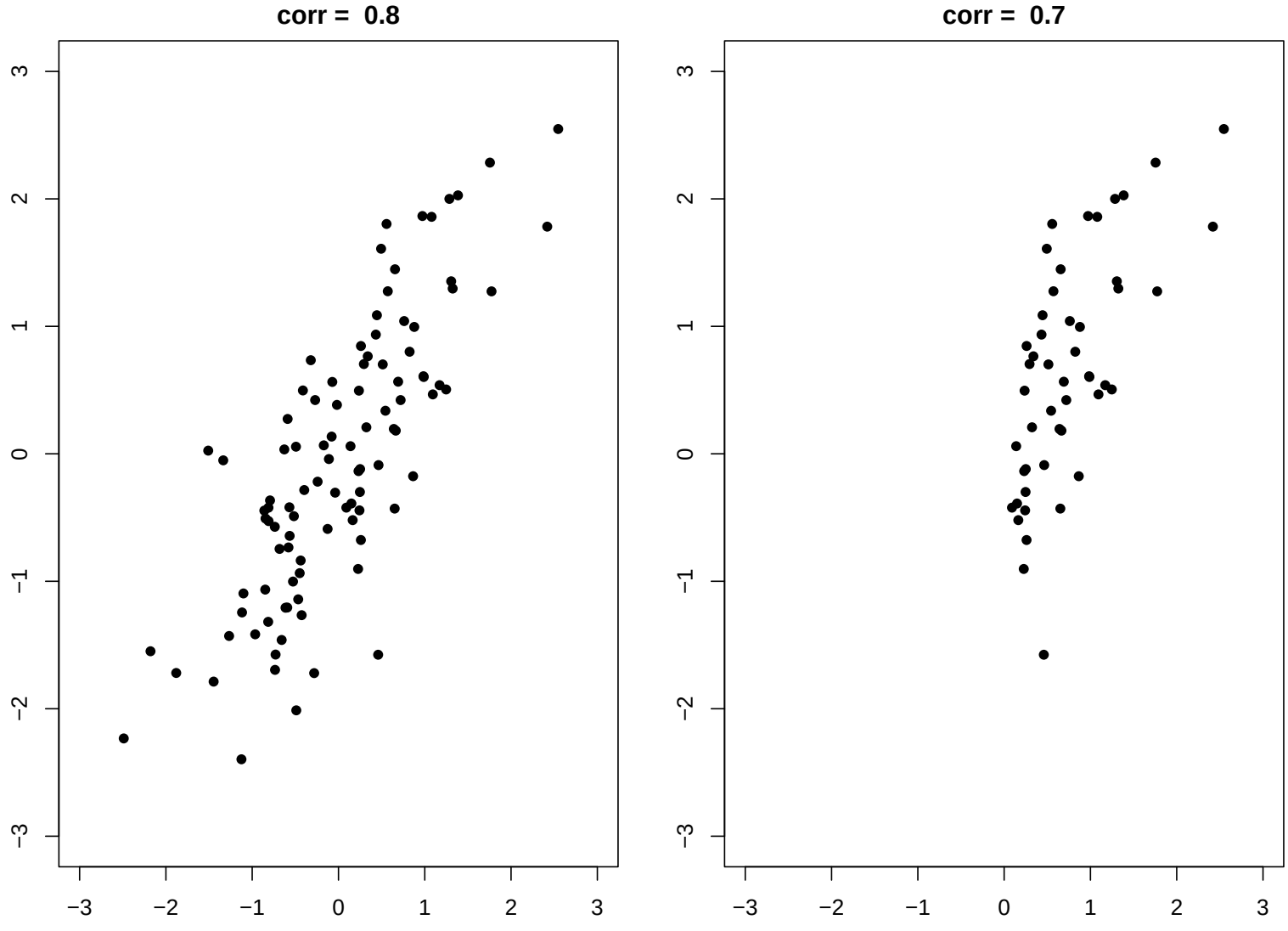


Shoe Size and Mathematical Ability

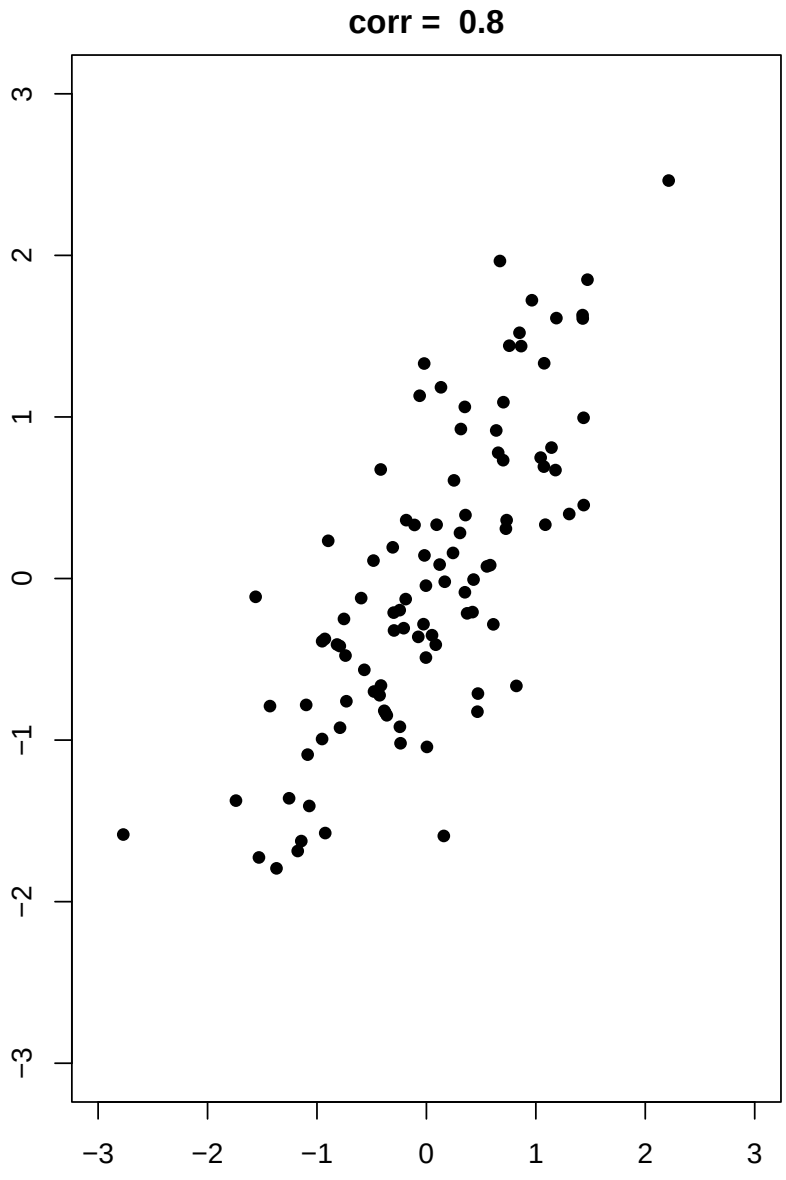
- What is the correlation of shoe size and mathematical ability among first graders?
- What is the correlation of shoe size and mathematical ability among fifth graders?
- What is the correlation of shoe size and mathematical ability among elementary school children?

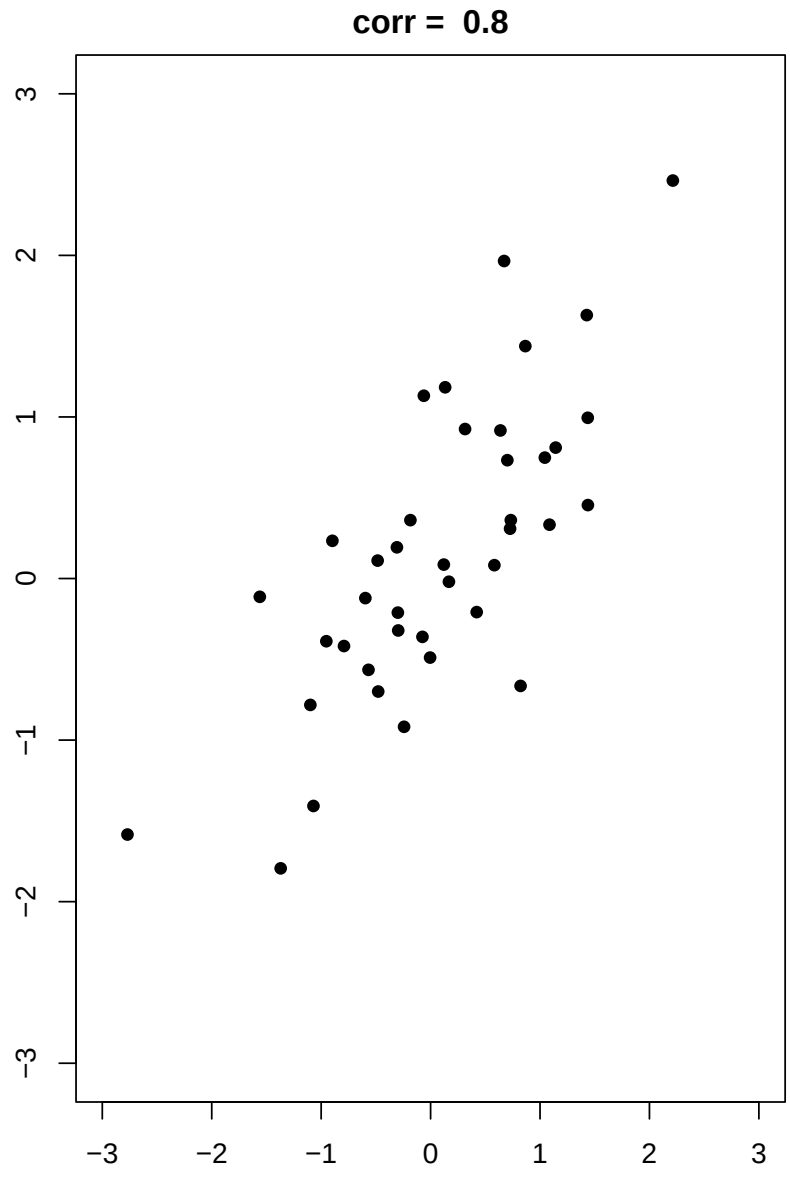
Taking a Subset 1

- This is sometimes referred to as **attenuation**.



Taking a Subset 2

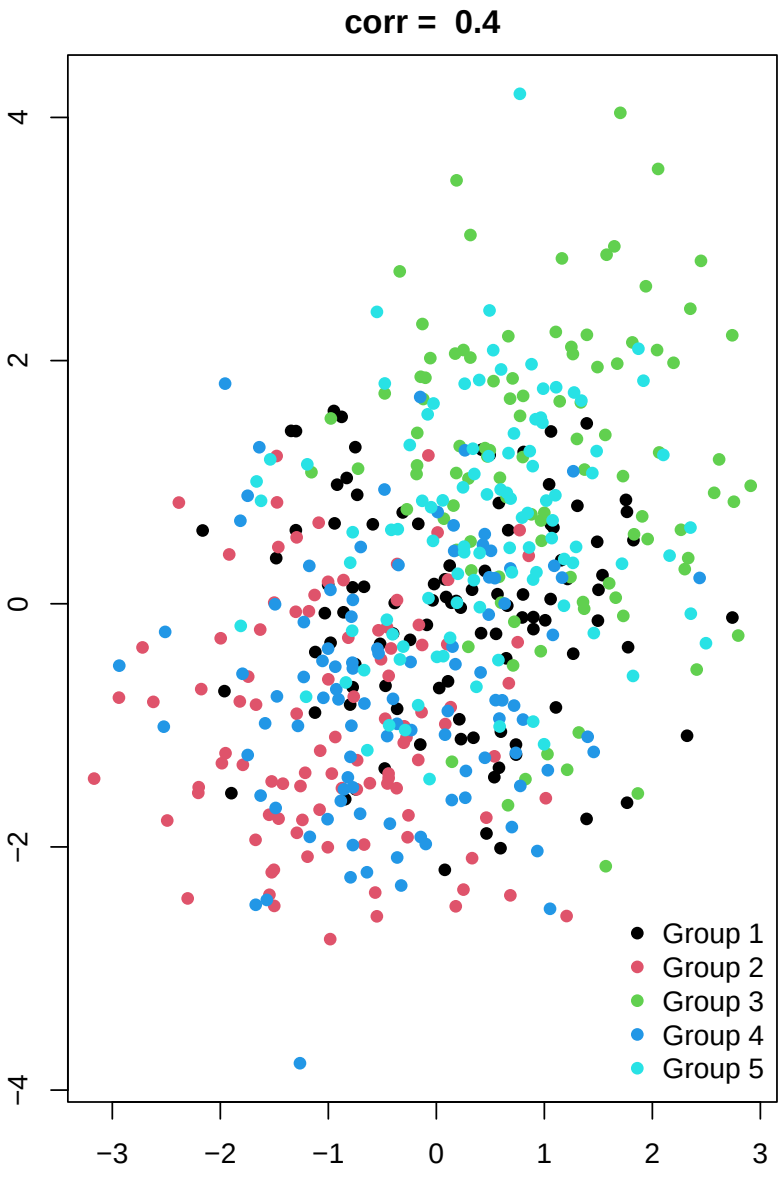


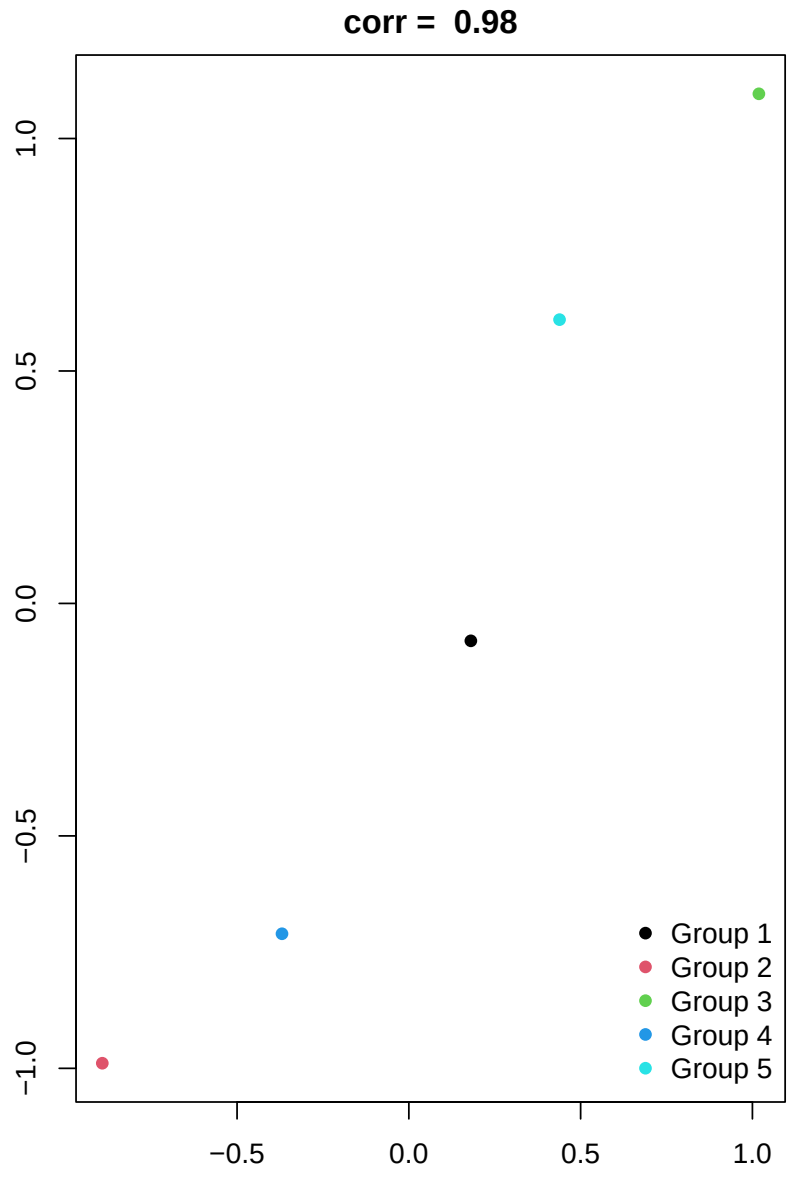


Height and Weight

- What is the correlation of height and weight among US adults?
- What is the correlation of height and weight among US adult men?
- What is the correlation of height and weight among US women?
- What is the correlation of height and weight among Michigan adults?

Taking an Average





Education and Income

- Suppose that data for individuals isn't available, but averages by state are.
- What would happen if you took the correlation of average education and average income across states?
- There is a name for a correlation of averages: **ecological correlation**.
- The ecological correlation will generally be higher than the correlation between the variables at the individual level.
- Sidenote: Is correlation a good measure of association between education and income, even at the individual level? Why or why not?

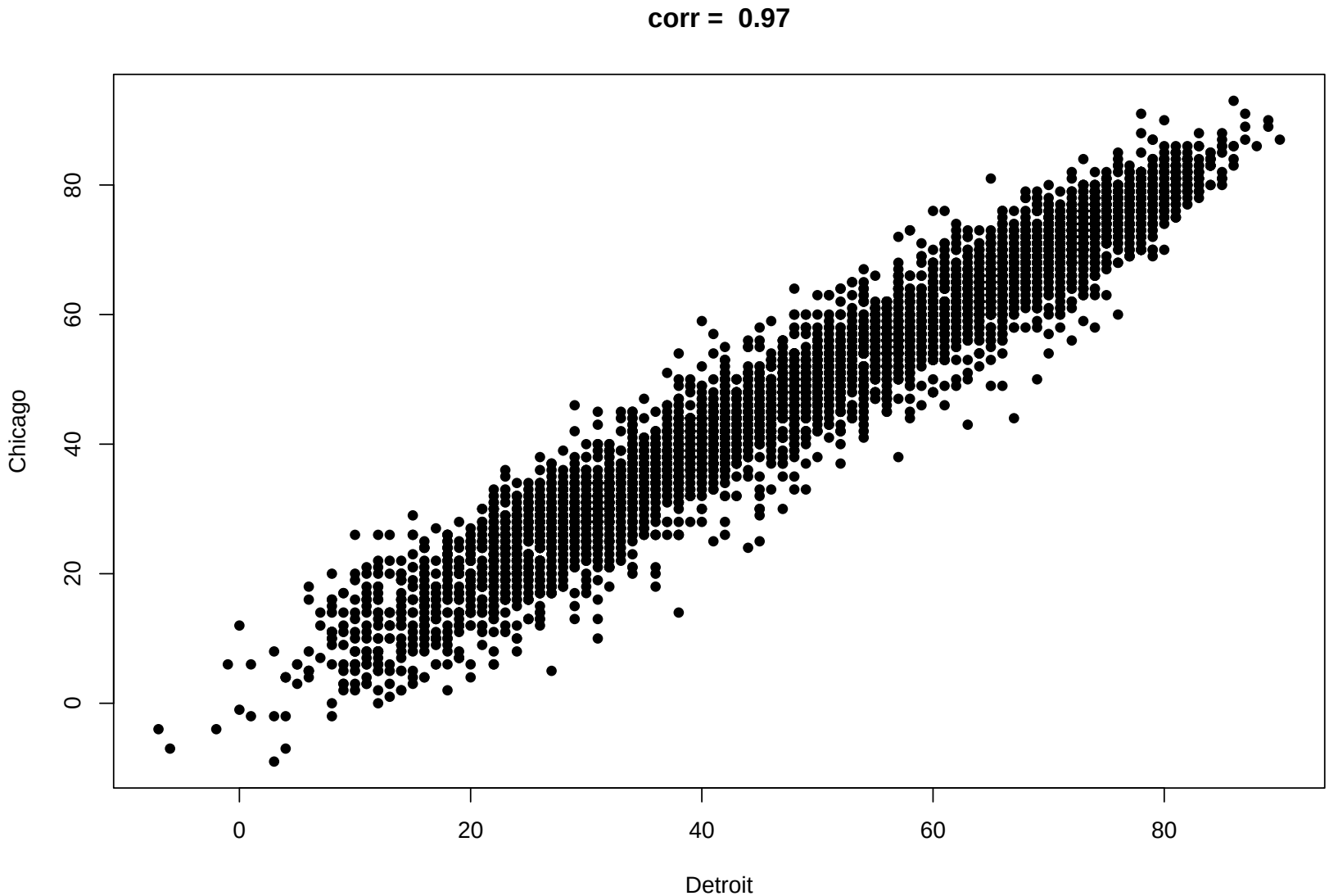
Conclusion

- Combining datasets can increase correlation if the relationships are similar and the datasets represent different ranges of x.
- Combining datasets can drive correlation toward 0 if the relationships are in different directions.
- Taking a subset can reduce correlation if the subset restricts (attenuates) the range of x.
- Taking averages reduces variability in both x and y and generally results in increased correlation.

It's a Correlated, Correlated World

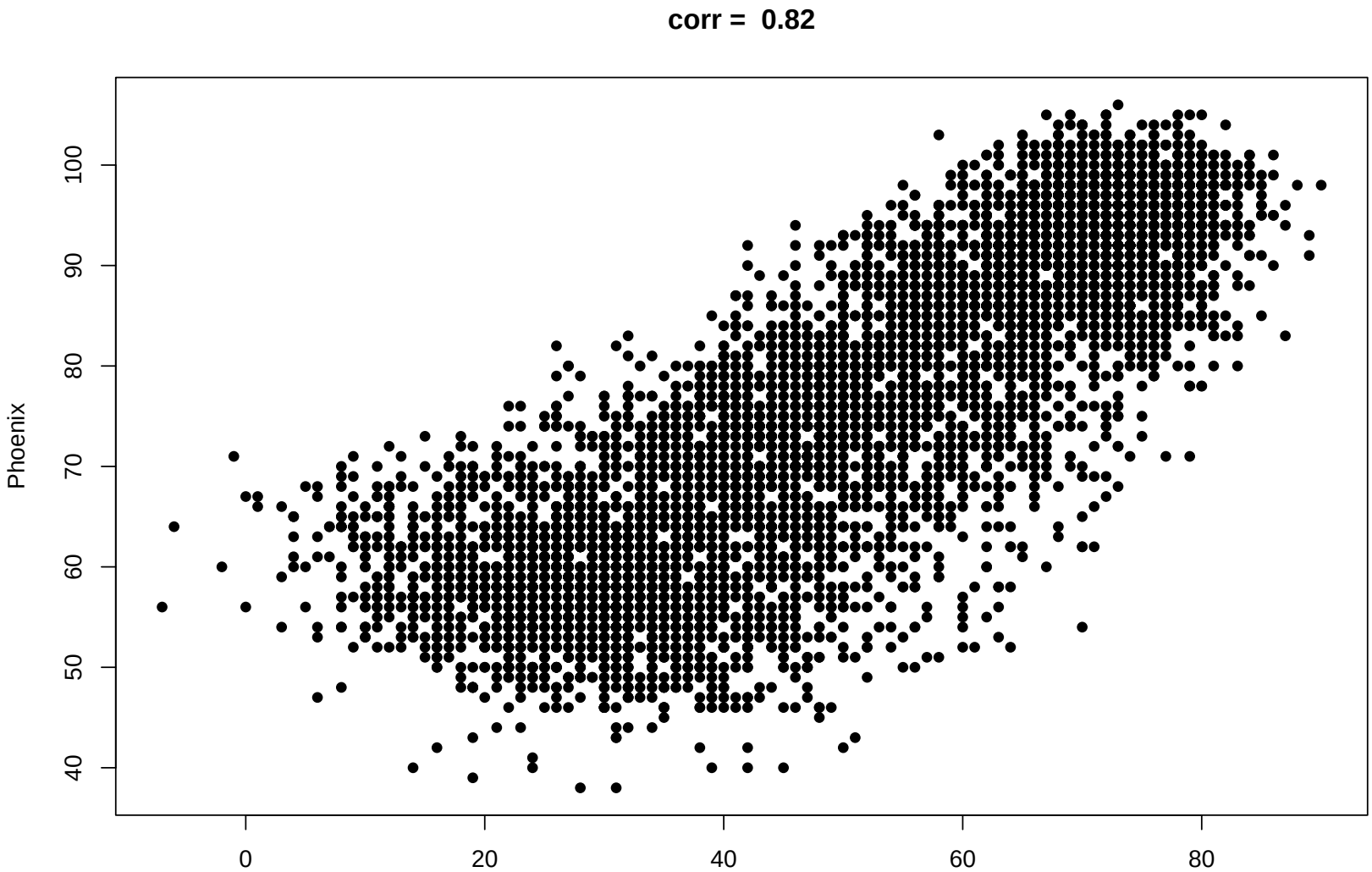
Temperature in Detroit and Chicago

- Daily average temperatures in Detroit and Chicago over the past 10 years



Temperature in Detroit and Phoenix

- Daily average temperatures in Detroit and Phoenix over the past 10 years
- Is correlation a good way to measure the relationship?



Temperature in Detroit and Sydney

- How have the average daily temperatures in Detroit and Sydney been correlated over the past 10 years?

Other Examples

- Do you think that these variables are correlated? Why or why not?
  1. Number of cellphones per capita in a country and the rate of breast cancer in the country
  2. Weekly ice cream sales and shark attacks
  3. Number of Nobel Prizes per capita in a country and the per capita consumption of chocolate

Nobel Prizes and Chocolate

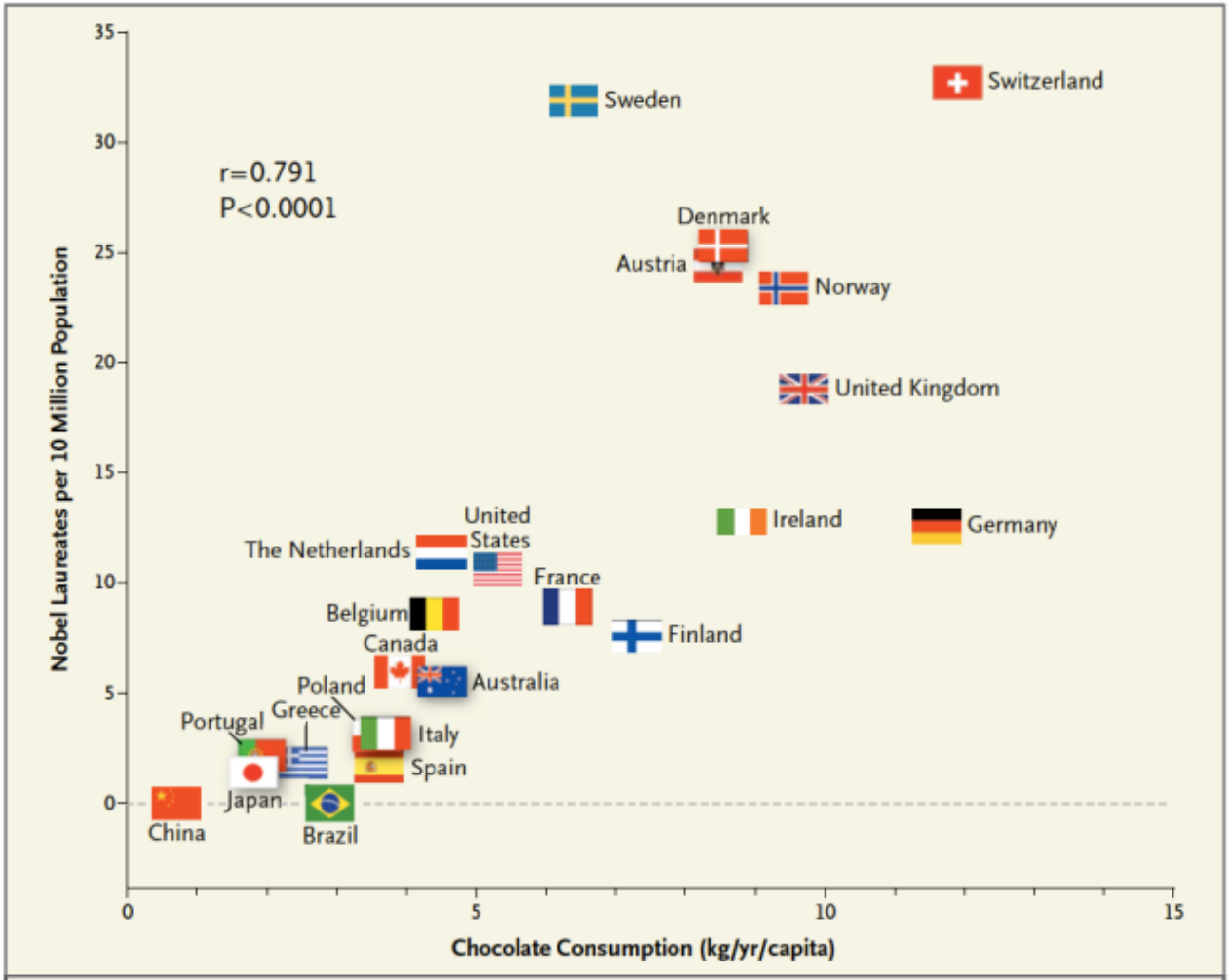


Figure 1. Correlation between Countries' Annual Per Capita Chocolate Consumption and the Number of Nobel Laureates per 10 Million Population.

Correlation is not causation!

Review Questions

- See Review_questions5-6.pdf